APIWAT NONUT

Developer Engineer | Industrial Software, Smart Factory Systems

Suratthani, Thailand

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GitHub: https://github.com/apiwatapply-svg

Professional Summary

Developer Engineer with 5+ years of experience across factory software, machine-side applications, smart factory dashboards, robotics automation, AI inspection, IoT systems, and deployment workflow. Strong at translating production requirements into working software: requirement analysis, system architecture, database design, backend APIs, realtime dashboards, Git workflow, CI/CD, and PM2/on-premise deployment.

My strongest fit is factory-facing development where software must connect with real machines, operators, production supervisors, maintenance teams, and management reports. I can work from shop-floor pain points to deployable systems: machine monitoring, OEE, NG traceability, machine status, paperless workflows, operator screens, and automated release processes.

Target Role

Developer Engineer / Industrial Software Engineer / Smart Factory Developer

Focus areas:

- Factory web applications and on-premise production systems
- Machine data flow: PLC / machine signal, MQTT, InfluxDB, SQL Server, realtime dashboard
- OEE, output, target, downtime, NG, status, alarm, and production report systems
- Machine-side deployment with PM2, Git workflow, CI/CD, and release documentation
- AI machine vision inspection using Python, camera, lighting, and SQL logging

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Core Technical Skills

Frontend: React.js, Next.js, HTML/CSS, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, Python, FastAPI, Django, Flask

Database: SQL Server, PostgreSQL, Supabase, Prisma ORM, ORM concepts

Factory / Realtime: MQTT, InfluxDB, Socket.IO, machine status, OEE, NG, alarm data

DevOps: Git, GitHub Actions, CI/CD, YAML pipeline, PM2, Linux server, deployment scripts, Docker basics

AI / Automation: OpenCV, YOLO, PyTorch, machine vision, photometric stereo, ROS, ROSBridge, AGV control

Tools: Figma, GitHub, AI Agent tools

Professional Experience

Developer Engineer

NMB-Minebea Thai Ltd.

Aug 2023 - Present

- Developed smart factory software and machine-side applications for production operation, monitoring, automation, and paperless workflows.
- Built and documented real-time Machine Monitoring System (MMS) workflows, including machine status, output, target, OEE, NG, alarm, and daily report views.
- Designed machine data architecture using MQTT, InfluxDB, SQL Server, Node.js backend APIs, Socket.IO realtime updates, and Next.js dashboard screens.
- Created Git workflow, CI checks, deployment documentation, PM2 process control, and release steps for factory software delivery.
- Developed ROS-based automation systems such as AGV and robot-related control workflows, combining ROS navigation, web dashboard control, and field test evidence.
- Worked on AI-based machine vision inspection concepts using industrial camera workflows, lighting control, image processing, YOLO classification, and SQL traceability.
- Collaborated with production, maintenance, and factory users to translate operational pain points into practical software features.

IoT Engineer

Kubota Research & Development Asia

Sep 2021 - Aug 2023

- Applied IoT technologies to smart agricultural machinery, drone-related systems, and auto-steering concepts.
- Conducted feasibility studies, market and product requirement analysis, and performance testing for new product concepts.
- Validated smart machinery features through field testing and technical evaluation to improve product reliability and customer trust.

Facilitator

Phoenix Pulp and Paper Plc Ltd.

May 2019 - Dec 2019

- Facilitated technical and soft-skill training for employees in a factory environment.
- Supported factory problem-solving activities and helped operators improve independent issue analysis.
- Built early experience in cross-functional communication, shop-floor support, and operational improvement.

Factory Projects

Smart Factory MMS Dashboard

Role: Full-stack Developer / Industrial Software Developer

GitHub: https://github.com/apiwatapply-svg/MMS_project

Tech: Next.js, React, Node.js, Express, SQL Server, Prisma, MQTT, InfluxDB, Socket.IO, PM2, GitHub Actions

Built an on-premise machine monitoring and OEE dashboard for 207 production machines, 27 machine types, and 4 factory zones. The system turns machine-side data into realtime visibility and historical reports for production review.

Key contributions:

- Designed data flow from machine signal to dashboard: PLC / machine data, MQTT, InfluxDB, SQL Server, backend API, Socket.IO, and web dashboard.
- Built report views for output, target, OEE, availability, cycle time, machine status, alarm, and NG traceability.

- Created layout dashboard for factory-wide machine visibility, so users can locate running, downtime, plan stop, and no-data machines visually.
- Added daily report, machine report, machine detail, and NG report views using real dashboard screenshots and report data.
- Prepared PM2 deployment workflow, /api/health check, CI workflow, unit tests, and deployment documentation for customer-server operation.
- Fixed realtime machine status flow so MQTT status updates insert into SQL Server and broadcast to the web immediately.

Business impact:

- Previous workflow required operators to record machine values manually. With 207 machines at around 1 minute per machine, one full round was approximately **207 minutes**.
- At a 25-minute recording interval, full manual recording would create up to **19 checks per 8-hour shift** and more than **65 manual hours per shift** if every machine had to be recorded one by one.
- New dashboard workflow reduces the need for repeated machine walking/checking by allowing users to monitor **207 machines / 4 zones** from one screen.
- Reduces manual paper/spreadsheet recording effort, lowers transcription error risk, and shortens response time from next-report review to **same-shift realtime exception monitoring**.
- Supports paperless reporting and creates searchable history for output, target, OEE, NG, status, alarm, and daily reports.

<u>Factory IoT Data Acquisition Device</u>

Role: IoT / Machine-side Developer

Tech: Arduino, machine signal input, MQTT, Node.js, InfluxDB, SQL Server, realtime dashboard integration

Developed IoT device concepts for collecting factory machine signals and sending machine-side data into digital monitoring systems.

Key contributions:

- Designed a machine-side data collection approach using Arduino to read machine signals/status from production equipment.
- Connected IoT data flow with MQTT/backend services so machine events can be used by dashboard and reporting systems.
- Supported the transition from manual machine checking to automated data collection for status, output, runtime, and abnormal condition visibility.
- Worked with factory requirements where devices must be simple, maintainable, and suitable for on-premise production environments.

Factory value:

- Reduces manual machine checking from repeated operator visits to automated signal capture at the machine side.
- Shortens data collection delay from manual 25-minute checking cycles to near-realtime MQTT/backend updates.
- Helps reuse existing machines by adding an IoT data layer instead of replacing machine control systems, reducing hardware change cost and implementation risk.
- Provides a scalable pattern: **1 device/data point per machine or process point** can feed dashboards, reports, and future OEE/maintenance analysis.

<u>Smart Factory Operations Platform</u>

Role: Full-stack Developer / Factory Maintenance System Developer

Tech: Next.js, React, Node.js, Express.js, Prisma, SQL Server, Socket.IO, Node-cron, Nodemailer, Bootstrap, Tailwind CSS, Axios, jsPDF, XLSX

Built a factory maintenance management system for machine preventive maintenance planning, maintenance history tracking, dashboard monitoring, user access control, and maintenance reporting.

Key contributions:

- Designed maintenance workflows for machine master data, machine type management, preventive maintenance planning, PM result recording, and maintenance history.
- Developed REST APIs with Express.js, Prisma, and SQL Server to support structured maintenance data instead of paper or spreadsheet tracking.
- Built Next.js/React screens for dashboard views, machine management, PM planning, PM results, and report pages.

- Added real-time machine status updates using Socket.IO for faster visibility across maintenance and production teams.
- Implemented scheduled PM notification logic using Node-cron and email notification support with Nodemailer.
- Added report export support for Excel/PDF to make maintenance records easier to review, share, and audit.
- Organized project documentation, environment setup, database scripts, and handover structure for deployment and future maintenance.

Factory value:

- Helps maintenance teams track upcoming PM tasks, completed PM results, machine history, MMS Dashboard, and repeated issues in one system.
- Reduces missed preventive maintenance activities and improves traceability of maintenance work.
- Supports paperless maintenance operations and creates a foundation for future integration with MMS/OEE data and predictive maintenance.

AI Defect Inspection System

Role: Machine Vision / AI Developer

GitHub: **https://github.com/apiwatapply-svg/Defect_Inspection_System**

Tech: Python, OpenCV, Ultralytics YOLO, PyTorch, Basler Camera, Photometric Stereo, SQL Server

Built an inspection prototype using a Basler camera, controlled blue lighting, photometric stereo processing, ROI extraction, YOLO OK/NG classification, result image generation, and SQL Server logging.

Key contributions:

- Built camera capture workflow and controlled lighting setup for repeatable part inspection.
- Processed four directional-light images into photometric stereo normal-map features to reveal surface defects.
- Cropped circular ROI points and classified each point as OK / NG / undefined using YOLO.
- Saved raw images, normal-map images, ROI crops, prediction images, final result images, and SQL inspection records.
- Designed point-level traceability for inspection timestamp and result review.

Factory relevance:

- Shows ability to combine hardware, camera, lighting, AI model, image processing, and database traceability.
- Helps reduce human inspection variation and creates evidence for later quality review.

Booking Meeting Room

Role: Full-stack Developer

Tech: Web application, admin workflow, room data, booking form, timeline schedule, mock room data, generated realistic room images

Developed a booking system with login, dashboard, room schedule, booking modal, admin management, and 2-5 meeting room data.

Factory relevance:

- Demonstrates internal-service workflow design, resource scheduling, admin controls, and visual UI documentation.

Non-Factory Project

Coffee Shop POS System

Role: Full-stack Developer

GitHub: https://github.com/apiwatapply-svg/POS_coffee

Tech: Next.js, SQL Server, Prisma, server actions, protected routes, role-based access, PM2-ready deployment

Built a production-ready POS system with cashier checkout, live barista queue, product management, order history, receipt pages, dashboard metrics, and role-based workflows.

Factory relevance:

- Demonstrates operational web application design: short user flow, reliable data persistence, role-based access, and deployment planning.
- Similar software pattern can be adapted to shop-floor paperless forms, operator screens, and transaction-based production workflows.

DevOps / CI-CD Capability

- Designed practical Git workflow for factory software: coding, feature branch, pull request review, CI test/build, CD release, and PM2 deployment.
- Built and explained CI workflow for MMS: backend unit tests, frontend build gate, smoke health check, and GitHub Actions reporting.
- Prepared one-command local CI scripts and documented how a developer can run checks before pushing code.
- Created PM2 deployment approach for customer server: pull/update code, install dependencies, build, restart process, and verify /api/health.
- Understands on-premise factory deployment constraints: offline/controlled network, environment variables, restart control, rollback planning, and operation manual needs.

Education

Master of Engineering in Mechanical Engineering

Khon Kaen University

2019 - 2021

Thesis: System Identification and Inner-Loop Stability Control for Fixed-Wing UAVs using Meta-Heuristics

Thesis Evaluation: Excellent

Bachelor of Engineering in Electrical Engineering

Khon Kaen University

2015 - 2019

Senior Project: Ping Pong Robot using LabVIEW and NI myRIO

GPA: 3.35, Second Class Honors

Certifications

- Mastering Indoor Vertical Farming for Cannabis (6 Hrs)
- Test Case Design Techniques
- Basic STM32 Microcontroller Programmer
- KOBDEMY Full-stack Developer Program: Figma UI/UX, Python, SQL & ORM, React, Next.js, Django, Flask, FastAPI, Node.js & ORM, Java, Spring Boot, Docker, Final Integration Project & Portfolio

Languages

- Thai: Native
- English: Conversational